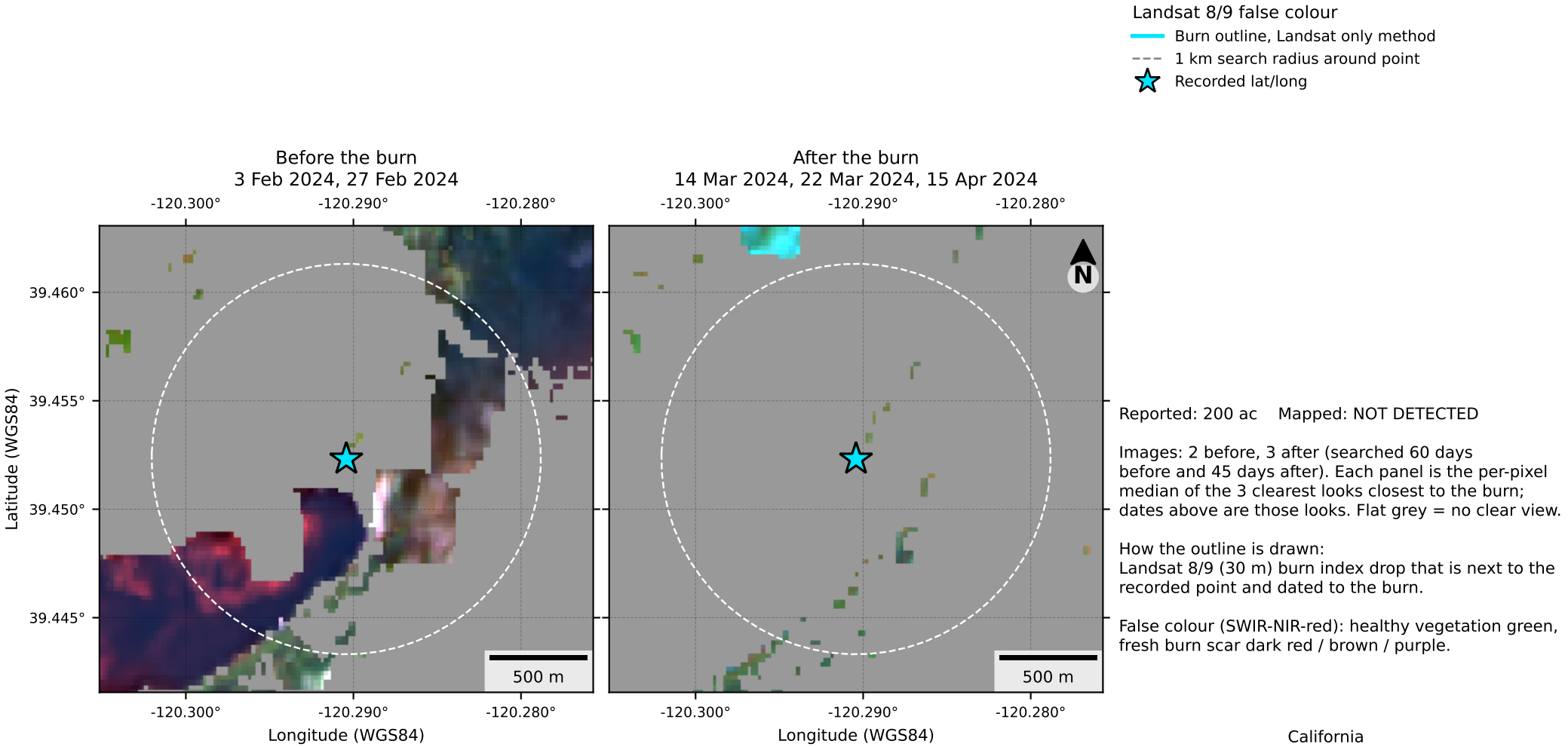
Independence Lake 2024 Rx Burn

Map 1 of 5: Landsat 8/9 before and after the burn | prescribed burn 6 Mar 2024 | reported 200 ac



Reported: 200 ac Mapped: NOT DETECTED

Images: 2 before, 3 after (searched 60 days before and 45 days after). Each panel is the per-pixel median of the 3 clearest looks closest to the burn; dates above are those looks. Flat grey = no clear view.

How the outline is drawn:
Landsat 8/9 (30 m) burn index drop that is next to the recorded point and dated to the burn.

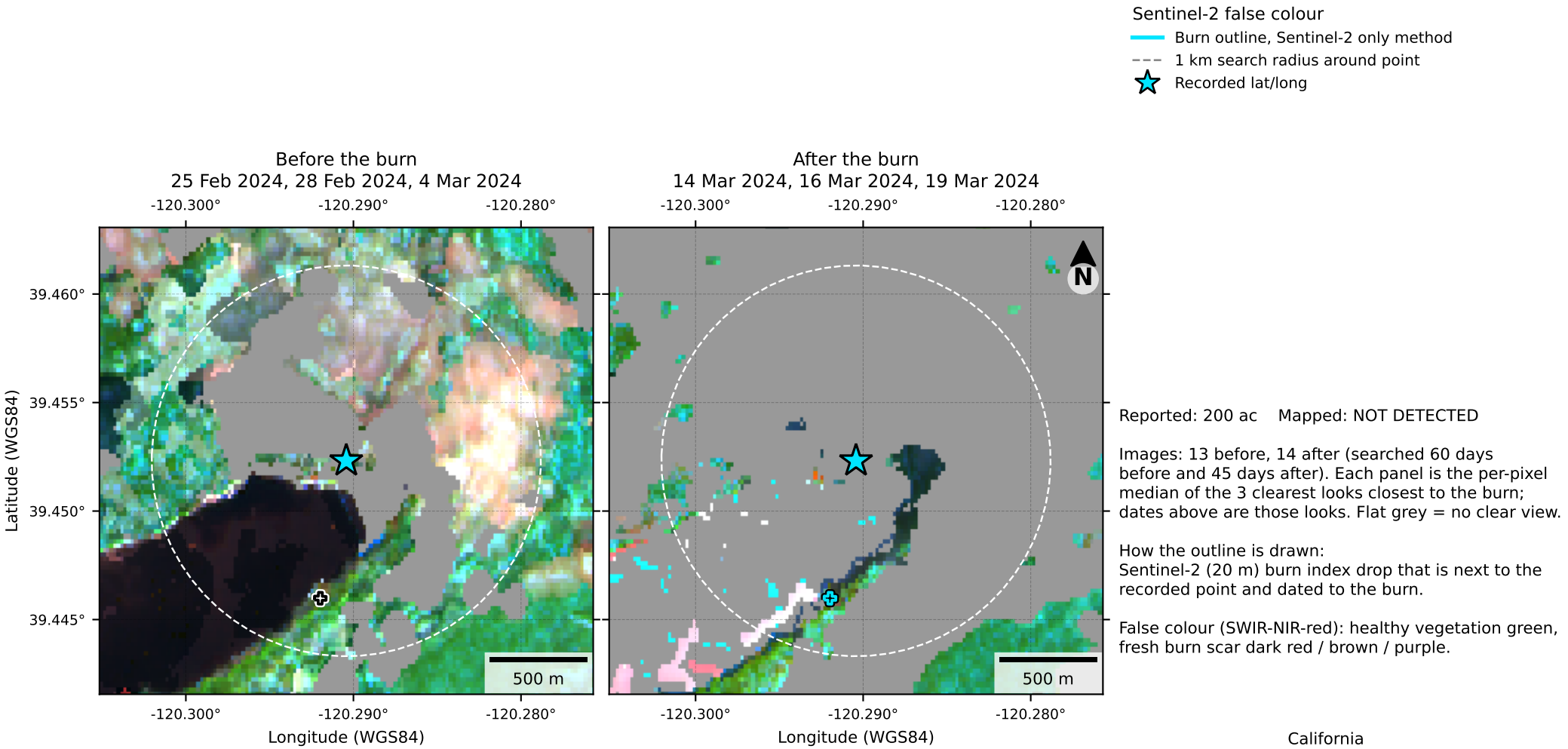
False colour (SWIR-NIR-red): healthy vegetation green, fresh burn scar dark red / brown / purple.



Satellite estimate from a recorded point, date and acres (free data); not field-verified. Prescribed burns under canopy often read as unburned even where surface fuels burned.
Landsat 8/9 surface reflectance (30 m resampled to 20 m). Coordinates WGS84 decimal degrees.

Independence Lake 2024 Rx Burn

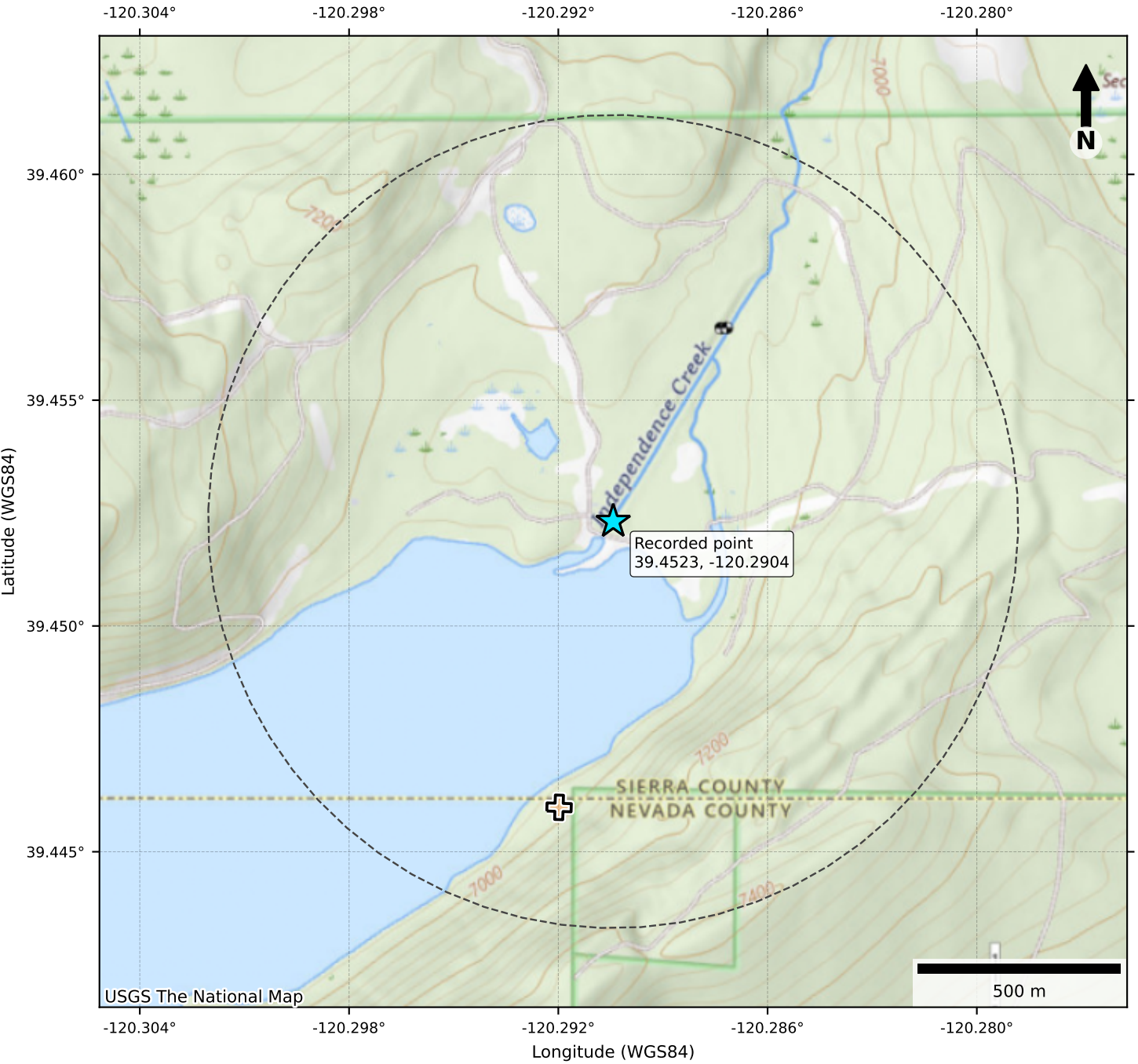
Map 2 of 5: Sentinel-2 before and after the burn | prescribed burn 6 Mar 2024 | reported 200 ac



Satellite estimate from a recorded point, date and acres (free data); not field-verified. Prescribed burns under canopy often read as unburned even where surface fuels burned.
Sentinel-2 surface reflectance (20 m). Coordinates WGS84 decimal degrees.

Independence Lake 2024 Rx Burn

Map 3 of 5: Sentinel-2 + Landsat combined | prescribed burn 6 Mar 2024 | reported 200 ac



Combined method

- Mapped burned area
- 1 km search radius around point
- Recorded lat/long

Reported: 200 ac Mapped: NOT DETECTED
Images: 13 + 14 Sentinel-2, 2 + 3 Landsat (before + afte
First clear image after burn: 8 days

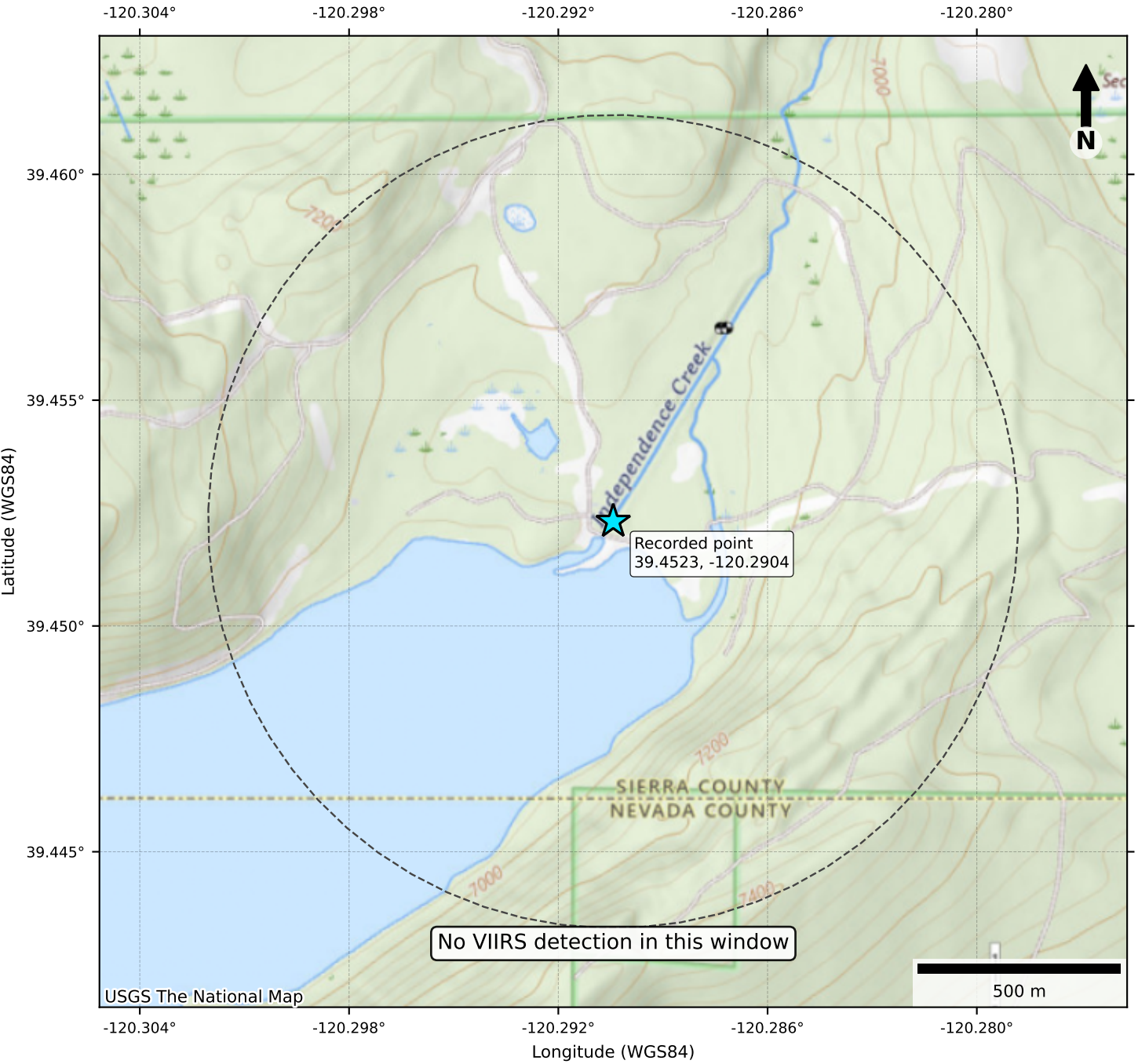
How the outline is drawn:
Both satellites in one time series (more clear looks).
Each image is compared with land far from the point,
so seasonal change cancels out. No embeddings.



Satellite estimate from a recorded point, date and acres (free data); not field-verified. Prescribed burns under canopy often read as unburned even where surface fuels burned.
Combined Sentinel-2 + Landsat burn index, point + date match. Coordinates WGS84 decimal degrees.

Independence Lake 2024 Rx Burn

Map 4 of 5: VIIRS active-fire detections (NASA FIRMS) | prescribed burn 6 Mar 2024 | reported 200 ac



VIIRS active fire

- VIIRS fire detection (375 m pixel footprint)
- 1 km search radius around point
- Recorded lat/long

Window: 5 Mar - 8 Mar 2024
Detections within 1 km of the point (+ half a pixel): 0
All detections loaded: 0 detections within ~30 km

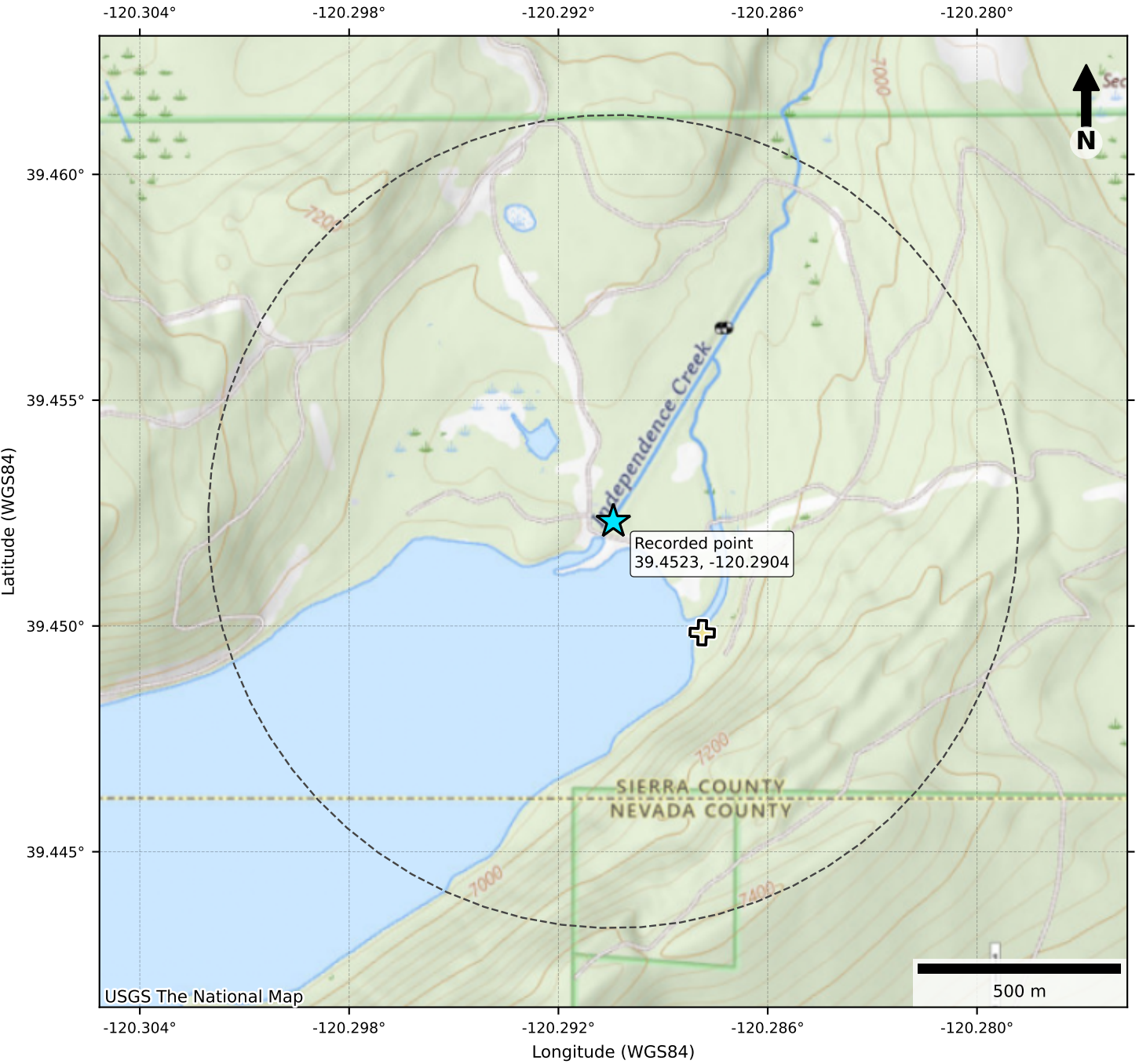
VIIRS (Suomi-NPP / NOAA-20 / NOAA-21) sees heat from active flames when a satellite passes (about twice a day per satellite). It confirms fire was present, not how far it spread; small, cool or short burns are often missed.



Satellite estimate from a recorded point, date and acres (free data); not field-verified. Prescribed burns under canopy often read as unburned even where surface fuels burned.
VIIRS 375 m active-fire product, NASA FIRMS. Coordinates WGS84 decimal degrees.

Independence Lake 2024 Rx Burn

Map 5 of 5: best effort (adds embeddings) | prescribed burn 6 Mar 2024 | reported 200 ac



Burn severity (satellite index)

- Unburned / very low (dNBR < 0.10): 0 ac
- Low (0.10 - 0.27): 0 ac
- Moderate-low (0.27 - 0.44): 0 ac
- Moderate-high (0.44 - 0.66): 0 ac
- High (> 0.66): 0 ac
- Mapped burn perimeter
- 1 km search radius around point
- Recorded lat/long

Reported: 200 ac Mapped: NOT DETECTED
Images: 28 Sentinel-2 + 5 Landsat
First clear image after burn: 8 days
Confidence: LOW
  - not detected (< 1 ha mapped)

How the outline is drawn:
As the combined method, but each pixel is compared only with look-alike land (same vegetation type, from TESSERA satellite embeddings of the year before), which removes more green-up / dry-down noise.



Satellite estimate from a recorded point, date and acres (free data); not field-verified. Prescribed burns under canopy often read as unburned even where surface fuels burned.
Severity = green-up-corrected dNBR, Key & Benson (2006) classes, not calibrated to field (CBI) plots. Coordinates WGS84 decimal degrees.